Delivery Performance, Delay Risk and Logistics Efficiency

A descriptive shipment-line case study using public / synthetic supply-chain data

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Status: Unpublished project technical report. Public hosting does not establish academic approval, peer review or journal publication.

Abstract

This project examines 180,519 order-item shipment lines across five markets, 23 regions and four shipping modes. It defines early, exactly on-time and delayed lines using actual minus scheduled shipping days, compares an existing late-delivery flag with that calculation, and presents mode and regional summaries. The browser companion uses an aggregated, privacy-minimized export of the original Python pipeline. The findings describe the supplied educational dataset; they do not establish real company performance or causal delay drivers.

Headline dataset results

Shipment lines	180,519
Within scheduled time (early + on-time)	42.72%
Calculated delayed lines	103,400 (57.28%)
Mean actual-minus-scheduled gap	0.566 days
Mean positive gap among delayed lines	1.617 days
Existing late-delivery flag	98,977 lines (54.83%)
Recorded sales	USD 36,784,734.31

1. Question and reporting unit

The analysis asks how delivery timing differs across shipping modes, markets and regions, and whether the existing risk flag agrees with a gap-based classification. One row is an order-item shipment line. There is no order identifier supporting an independent order-level denominator, and there is no date column supporting trends or seasonality.

2. Data preparation

The repository loads the release CSV using latin-1 encoding, strips text whitespace, standardizes the documented EE. UU. customer-country spelling, validates numeric/risk domains and derives the gap and cancellation flag. Canceled records remain in the data. Missing non-KPI customer attributes are not silently imputed. The browser export retains only mode, region, market, segment, gap, flag, cancellation status, counts and summed sales; it excludes names, addresses, customer IDs and coordinates.

3. Definitions

Gap = actual shipping days minus scheduled shipping days. A negative gap is early, zero is exactly on time, and a positive gap is delayed. The primary within-schedule rate combines early and exactly on-time lines. The mean gap includes negative values; the mean positive delay is a separate measure. The existing Late_delivery_risk field is a dataset attribute, not a new prospective prediction.

4. Results and interpretation

First Class has a 100% positive-gap rate in the supplied data. This is a reason to investigate the relationship between the mode labels and promised durations, not proof of a real carrier failure. Mode-level delayed percentages differ more than the reported high-volume regional percentages, but those descriptive differences do not isolate causal effects. The risk flag aligns with the dataset delivery-status label; 4,423 differences from the gap-based classification relate to canceled lines. A cancellation sensitivity view is available rather than assuming canceled records represent observed completed deliveries.

5. Financial interpretation

Approximately USD 21 million of sales is associated with delayed lines. It is exposure, not confirmed loss, penalties, refunds or savings. The dataset does not measure those outcomes, so financial recommendations require separate evidence.

6. Demonstration and reproducibility

The original Streamlit application remains at dashboard/app.py. The browser companion has four sections, dimension filters, a cancellation-basis toggle and summary CSV downloads. scripts/build_web_data.py recreates its aggregate export using the existing cleaning and KPI modules. The repository contains SQL, tests and generated reports. The hosted demo is a static client-side application, not a hosted Python server.

7. Limitations and next validation

The supplied dataset is a public / synthetic educational case, not independently verified APL Logistics operational data. There is no longitudinal evaluation, distinct-order key, causal experiment, observed carrier cost or penalty outcome. Before operational use, confirm provenance and rights, obtain true service-level definitions and evaluate whether segment/mode mix explains the descriptive differences.

Sources and reproduction

Source repository: https://github.com/rishi-1603/Delivery-Performance-Delay-Risk-and-Logistics-Efficiency-Analysis-in-Global-Supply-Chain-Operations

See docs/methodology.md, docs/kpi_dictionary.md, reports/analysis_report.md and the source cleaning/KPI modules. Original dataset provenance is documented in the repository; no external research approval is implied.